BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

SQL: SELECT, FROM and WHERE

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why close a file after use?

2. Which link best suits Bluetooth?

A) Nearby wireless keyboard
B) School buildings far apart
C) Undersea Internet cable
D) Long-distance fibre backbone

3. "Brute force works only by physically breaking the computer." Correct this. Explain why.

4. A network shows 2001:db8::7. "IPv6 uses four decimal numbers separated by dots." Correct this. How is this address written?

Today's objectives

Write simple SQL queries to retrieve records. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

SELECT
Specifies columns required from a database table.

FROM
Identifies the database table from which records are selected.

WHERE
Filters records using a condition.

SQL query
SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

SELECT	Specifies columns required from a database table. In the school club database, identify where this technique applies.
FROM	Identifies the database table from which records are selected. In the library books database, identify where this technique applies.
WHERE	Filters records using a condition. In the sports team records, identify where this technique applies.
SQL query	SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. In the museum member list, identify where this technique applies.

I Do - One worked example

A school club database uses select and from. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. SELECT [1]; Specifies columns required from a database table. [1].

3. FROM [1]; Identifies the database table from which records are selected. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library books database needs select. Explain the purpose. [2 marks]

2. Explain from for a sports team records. [2 marks]

3. When programming a museum member list, explain why where and sql query matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete this SQL query to return names of pupils with a mark greater than 60.

```
SELECT Name FROM Pupils WHERE Mark ... 60
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain from and where in a café inventory. [4 marks]

Q4. A programmer makes a school house records. Explain the roles of select, where and sql query. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. The SQL SELECT clause specifies which columns are retrieved from a database table. ☐ T ☐ F
2. Specifies columns required from a database table ☐ T ☐ F
3. Identifies the database table from which records are selected ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It releases the handle or resources and helps ensure buffered writes are completed. ☐

Q2 A - Nearby wireless keyboard ☐

Q3 It tries many possible passwords or keys until one works. ☐

Q4 IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN. ☐

Code First - Model

```
SELECT Name FROM Pupils WHERE Mark > 60
```

Output: Ada, Kai (example Pupils rows: Ada 62; Ben 45; Kai 80)

We Do - Check the model

1. SELECT [1]; Specifies columns required from a database table. [1]. ☐

2. FROM [1]; Identifies the database table from which records are selected. [1]. ☐

3. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 SELECT [1]; FROM [1]. ☐

Q3 FROM [1]; Identifies the database table from which records are selected. [1]. WHERE [1]; Filters records using a condition. [1]. ☐

Q4 SELECT [1]; Specifies columns required from a database table. [1]. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: